

PAW/USPP consolidation plan: D matrix, converters, matrix elements, ERIs

2026-07-24 (status: PROPOSED)

Abstract

Companion to `notes/paw.implementation.plan.md` (the phased build-out, now historical). This is the *corrective* plan: it defines the target design for PAW/USPP in the self-consistent GW framework, records the audit findings that motivate it, and sequences the work. All file:line references are against the `paw` branch working tree on 2026-07-24. The guiding requirements: no DFT contributions in the non-local D matrix (except a future, optional DFT- V_{xc} core–valence term); a clean separation between static (density-independent, built once at initialization) and dynamic (density-matrix-dependent, built at matrix-element evaluation) terms; removal of the QE `deeq` tensor; frozen core–valence exact exchange read from file and added to the bare D ; QE/ABINIT backend parity; both moment-restored and shape-restored augmentation densities available and correct; and, as the end goal, user-selectable evaluation routes for the Hartree and exchange contributions to the HF/GW static self-energies — factorized ERIs (THC/Cholesky, today’s only route) vs direct evaluation through the `hamiltonian` routines — which requires the two routes to be a compatible partitioning of the same matrix elements.

Contents

1	Design invariants (the contract everything below enforces)	2
2	Audit findings (what is wrong today)	4
2.1	pseudopot / matrix elements	4
2.2	Converters	5
2.3	Augmentation modes (moment vs shape)	6
2.4	Notes	6
3	Workstream A — pseudopot D-matrix refactor (first task)	6
4	Workstream B — converter parity	8
5	Workstream C — augmentation-density modes (moment & shape)	8
6	Workstream D — ERI/THC consistency (second task, after A–C)	9
7	Workstream E — notes and documentation	9
8	Sequencing and risk	10

1 Design invariants (the contract everything below enforces)

For every augmented species (USPP and PAW), with projector channels I, J on atom a of species $t(a)$, and CoQuí's self-consistent density matrix $n_{s,\mathbf{k},ab}$ (never `mf.occ()`):

I1. One-center density matrix (becsum)

A *single* becsum definition used by all consumers (Hartree, deeq, exchange, energies), symmetry-correct on any k -mesh:

$$\text{becsum}_a(I, J) = \frac{n_s}{N_k} \sum_s \sum_{\mathbf{k} \in \text{full BZ}} \sum_{ab} P_{a,aI}^{\mathbf{k}*} n_{s,\mathbf{k},ab} P_{b,aJ}^{\mathbf{k}}, \quad n_s = 2 \quad (n_{\text{spin}}=1, n_{\text{pol}}=1), \quad (1)$$

with the full-BZ lift of the projections P via the existing atom-permutation + Wigner- D + Bloch-phase machinery whenever $N_k^{\text{IBZ}} < N_k$.

I2. Static (frozen) per-atom D — built once at pseudopot initialization, never mutated

$$D_a^{\text{static}}(I, J) = D_{t(a)}^0(I, J) + \alpha_x E_{t(a)}^{cv}(I, J) \quad [+ \text{future: DFT-}V_{xc} \text{ c-v term}]. \quad (2)$$

- D^0 is the *full* frozen ionic d_{ij}^0 :

$$D_{IJ}^0 = (T^{\text{AE}} - T^{\text{PS}})_{IJ} + \langle \phi_I \phi_J | v_H[n_{Zc}] \rangle - \langle \tilde{\phi}_I \tilde{\phi}_J + \hat{Q}_{IJ} | v_H[\tilde{n}_{Zc}] \rangle, \quad (3)$$

i.e. AE–PS kinetic difference plus the frozen-core–screened ionic Hartree terms (including the compensation-moment piece on the PS side). QE's UPF *dion* is this object by convention; the ABINIT converter assembles it (`assemble_dij0: kinetic_energy_differences + v_H[n_Zc]/v_H[\tilde{n}_{Zc}]` matrix elements + moment term). CoQuí does *not* re-derive D^0 when the `h5` provides it; `compute_paw_static_D` remains available as a cross-check tool only.

- E^{cv} = frozen core–valence exact exchange (`ex_cvij`), contracted with **factor 1** — it is linear in the valence density matrix and is captured through $e_{1e} = \text{Tr}[D_m H_0]$, *never* through $\frac{1}{2} \text{Tr}[D_m K]$. Source order: `Onecenter/ex_cvij` if present; else built natively from `Core/ae_wfc` (§4, item B3); else omitted with a loud warning.
- The DFT- V_{xc} core–valence option (a DFT c–v XC term added to the bare D) remains a declared-but-aborting enum value (`cv_exchange_e::dft_xc`) until implemented.
- No DFT XC and no QE-density quantities anywhere in D^{static} .

I3. Dynamic (density-dependent) per-atom D — Hartree only, rebuilt at every matrix-element evaluation from the current n

$$D_a^H(I, J)[n] = \int d^3r v_H[\bar{\rho}[n]](\mathbf{r}) \hat{Q}_{a,IJ}(\mathbf{r}) + \Delta D_a^H(I, J)[\text{becsum}[n]], \quad (4)$$

where $\bar{\rho}$ = smooth density + compensation charge, and ΔD^H is the radial AE–PS one-center Hartree (`compute_paw_hartree_atom`; proper Hartree units, `rad_fac` = 1, frozen-core AE/PS core

densities included). For the full KS-like operator the $\int V \hat{Q}$ piece uses $V_{\text{loc}} + V_H$; for the pure Hartree matrix it uses V_H only. Both already exist in `compute_paw_deeq_from_becsum`; they become the *only* source of density-dependent D .

I4. Exchange never enters D

Valence–valence exact exchange lives in v_x / the THC ERI, with two selectable augmentation-density modes:

- **moment** (default): moment-restored compensation charge $\hat{Q}$ in the pair densities *plus* the ΔC one-center correction, prefactor $s_{\text{oc}} = -1/N_k$. This is the Fock-exact-one-center operator (ABINIT `pawdijfock` / VASP analogue).
- **shape**: full AE–PS partial-wave pair density ($\phi\phi - \tilde{\phi}\tilde{\phi}$ form factors, `build_grad_tab_full_aeps`), ΔC *dropped* (double-count otherwise), evaluated on the dense grid. This is the Arnaud / GW-oscillator operator.
- In *both* modes core–valence exchange comes from E^{cv} in D^{static} (Eq. 2).

Both modes must be available and numerically correct in the direct v_x path *and* the THC ERI path. Hartree always uses moment restoration (compensated smooth + radial one-center is already exact for Hartree — document, don’t option-ize).

I5. H_0 and evaluation-path semantics

$$H_0 = T + V_{\text{loc}}(\text{pp_local}) + \sum_{a,IJ} |\beta_{aI}\rangle D_a^{\text{static}}(I, J) \langle \beta_{aJ}|. \quad (5)$$

$e_{1e} = \text{Tr}[D_m H_0]$ is then purely static + frozen-core. The `nii/nij-vs-nullptr` asymmetry in `add_vpp_impl` is *by design* and serves route selectability (I7): density-dependent terms are added **if and only if** a density (`nii` or `nij`) is passed.

- **`nii = nij = nullptr`**: Eq. (5) only. Hartree and exchange are supplied elsewhere (today: by the ERI objects orchestrated in the SCF classes) and must *not* appear here — no QE `deeq`, no dynamic D .
- **density passed**: the requested density-dependent terms are added on top of Eq. (5), under *separate* boolean control for Hartree and for exchange (new `add_hartree` / `add_exchange` flags on `add_vpp_impl` and the public `add_Vpp` overloads), so the future `set_H` can mix routes — e.g. Hartree direct + exchange via THC.
- the `nii` and `nij` overloads must produce the identical operator for the same physical density.

QE-eigenvalue reproduction is a *diagnostic* that assembles $D^{\text{static}} + D^H[n_{\text{QE}}] + \int V_{xc} \hat{Q}$ explicitly — it is not a production path.

I6. Backend parity

A QE-converted and an ABINIT-converted h5 for the same system must drive the identical CoQuí code path, with no per-backend branches in `pseudopot` beyond dataset presence (`ex_cvij`, `Core/`). Same datasets, same units, same index conventions.

17. Evaluation-path equivalence (factorized ERIs vs direct)

Today the Hartree and exchange contributions to the HF and static-GW self-energies are built *exclusively* from ERI objects (THC or Cholesky), controlled in the SCF classes. The end goal is user choice: a future `set_H` routine will optionally assemble V_H and/or Σ_x directly from the `hamiltonian` routines (`add_Vpp` with density + flags, `add_Hartree`, `add_exchange`), with the ERI route remaining available. The two routes must therefore be a *compatible partitioning of the same Hamiltonian matrix elements*:

$$J_{ij}^{\text{ERI}}(s, \mathbf{k})[n] = J_{ij}^{\text{direct}}(s, \mathbf{k})[n], \quad K_{ij}^{\text{ERI}}(s, \mathbf{k})[n] = K_{ij}^{\text{direct}}(s, \mathbf{k})[n], \quad (6)$$

for every spin, k -point, and band pair — up to converged ISDF thresholds and compatible cutoffs. E.g. the HF build through the PAW-enabled THC object must reproduce, element by element, the `hamiltonian`-routine result. Eq. (6) is the acceptance criterion for Workstream D and the gate for the future route-selection project.

18. All-electron orbital basis; identity overlap

For PAW and USPP, HF and GW operate in the *all-electron* orbital basis: every matrix element (H_0 , V_H , Σ_x , ERIs) is taken with respect to the AE orbitals $|\psi_i\rangle = \hat{T}|\tilde{\psi}_i\rangle$, which are orthonormal — the overlap in all SCF/Dyson algebra is the **identity**. The smooth orbitals + projector augmentation are only the *representation* of these AE matrix elements; `nii`/`nij` are AE-basis density matrices, and the `becsum` of Eq. (1) is the AE one-center density matrix. No S -matrix insertions appear in any density contraction.

2 Audit findings (what is wrong today)

2.1 pseudopot / matrix elements

- F1. QE deeq is read and used as the production D .** `pseudopot.cpp:584` reads `deeq` into `Dnn_atom`; `QE deeq` = $d_{\text{van}} + \int V_{\text{eff}} \hat{Q} + \text{ddd_paw}$ with $V_{\text{eff}} \supset V_{xc}$, and `pw2coqui` refreshes `ddd_paw` from the converged DFT density before export (`pw2coqui.f90:213–219, 458–466`). DFT XC therefore enters H_0 twice over. This is the central violation; removal is directed.
- F2. The three `add_vpp` entry points are mutually inconsistent** for USPP/PAW: *no-density* (production `set_H0` for GW/scGW via `gen_one_body_hamiltonian.icc`): `QE deeq` + `ex_cvij` (`pseudopot.cpp:1308–1311`); `nii`: native `compute_paw_deeq(nii, V_{\text{loc}}+V_H, include\_static=true)` — no `QE deeq`, but also **no** `ex_cvij` (`pseudopot.cpp:1278–1293`); `nij`: falls through to `QE deeq` again (MAM FIX FIX FIX at `pseudopot.cpp:1277`, comment 1303–1307). Same density $\Rightarrow$ three different Hamiltonians. NB: the *inclusion* asymmetry — density-dependent terms only when a density is passed — is *intentional* (route selectability, I5/I7). The defects are that the tensors disagree: the no-density path must be static-only (not `QE deeq`), and `nij` must build the same native dynamic D as `nii`.
- F3. `ex_cvij` placement is split.** Added to `Dnn_atom` in the ctor (`pseudopot.cpp:987–1007`) but absent from `Dnn_atom_static_paw` (`paw_onecenter.hpp:388–413`), so (a) the native `nii` path drops c–v exchange, and (b) `compute_deeq_scf` does `Dnn_atom` = static + Hartree and would *erase* `ex_cvij` from `Dnn_atom` if ever called in production (`paw_onecenter.hpp:459–461`).

- F4. becsun symmetry is path-dependent.** The diagonal path uses the full-BZ `compute_becsum_diagonal_symm` all `nij` consumers (`v_h(nij)`, `compute_paw_deeq(nij)`, `compute_deeq_scf`) use the IBZ-weight `compute_becsum_full` (`v_h.paw.hpp:147-148`), and `add_Hartree(nij)` has no $N_k^{\text{IBZ}}=N_k$ guard (v_x 's `nij` overload does). Symmetry-reduced mesh + full $D_m \Rightarrow$ silently wrong compensation charge. `compute_becsum_full` also truncates the imaginary part with only a log (`v_h.paw.hpp:173-184`) instead of Hermitian pair symmetrization.
- F5. Static-ish work recomputed per call:** `aainit_tables_build`, `qgrad` tables, and the full-BZ Pskna lift are rebuilt inside every `compute_paw_deeq` / v_x call (`paw_onecenter.hpp:565`; `v_x.paw.hpp:397-413`, 375, 740); no Δk -keyed Qfac cache. Known perf sink (direct v_x went hours $\rightarrow$ 27s after the uncommitted fixes; caching is the next step).
- F6. Spin handling:** `deeq` spin-collapse to $s=0$ and `vfull = svloc(0,0,.)+v_r` (`pseudopot.cpp:600-603`, 1285-1286) — magnetic LSDA D is a documented TODO; the refactor must leave a clean seam (per-spin D^H) rather than bake $n_{\text{pol}}=1$ deeper in.
- F7. Dead/legacy surface:** unused member `memory::unified_array qq` (`pseudopot.h:471`); `svsc` read but unused in production paths; `skip_pp` / `set_kinetic` diagnostic surface; `compute_paw_static_D` unconsumed; stray `.paw_onecenter.hpp.swp`; no-op root check `pseudopot.cpp:699`. The `xml/pw2bgw` init path is explicitly out of scope (user directive) — leave untouched.

2.2 Converters

- F8. QE exports `deeq/deeq_nc`** (`pw2coqui.f90:463-466`) — to be deleted with F1. Everything else QE writes for the D chain (`dion`, Species radial data, `vloc_ps/rho_atc_ps`, `Onecenter/deltaC` in proper Ha) is XC-free and already sufficient for native reconstruction.
- F9. ABINIT h5 is missing datasets the native path needs:** Species/`paw/{ae_vloc, vloc_ps}` are computed internally ($v_H[n_{Zc}]$ / `reconstruct_vhtnzc`) but never written — so CoQui's static/dynamic rebuild cannot run on ABINIT h5 today; `vxc/vxc_with_nlcc` are written as *zeros* (G_0W_0 $\Sigma - V_{xc}$ broken by construction); `nuclear_energy` and `madelung_constant` are 0.0 (madelung patched reader-side); Species/`beta` absent; no Core/ group; parsed `exx_core_core` dropped.
- F10. Cross-backend convention mismatches:**
- `proj_per_atom`: QE writes per-SPECIES `nh(1:nsp)` (`pw2coqui.f90:429`); ABINIT writes a per-ATOM array of length n_{at} (`abinit_paw_hamiltonian.py:369-371`, 427). The reader sizes it to n_{sp} (`pseudopot.cpp:511-514`) — works only for ordering-degenerate cells (Si-type). Latent multi-species breakage.
 - Units: /Hamiltonian datasets in Ry (QE convention, reader $\times \frac{1}{2}$) but `Onecenter/*` in Ha; Species-level `dion` is Ry on QE, Ha on ABINIT. One standard (Ha on disk + schema version attribute) is required.
 - `miller_g`: `ecutrho` sphere (QE) vs full FFT box (ABINIT).
 - `qfuncl` provenance: UPF-tabulated (QE) vs moment $\times$ analytic Bessel shape (ABINIT), with the tabulated `<shape_function>` never cross-checked.
 - `ijtoh` leading dimension for modern QE (per-atom?) unverified.

F11.

`ex_cvij` exists only on the ABINIT path; QE-PAW datasets get `cv_exchange=none` + warning even when GIPAW Core/ae_wfc is exported (pw2coqui.f90:991-1005) and would suffice to build it.

2.3 Augmentation modes (moment vs shape)

F12.

Shape restoration works in direct v_x (nii and nij, ΔC correctly dropped) but: THC aborts on it (smooth-grid capture $\sim 6\%$ — needs the dense-grid augmentation); the direct path builds grad tables per call with $dq=0.05$ while THC uses $dq=0.01$; and *two* parallel flags (`paw_exx_shape_restored` bool + `_exx_opts.vv_compensation`) are kept in sync by hand (pseudopot.h:393-413). The shape-restored mode is not yet numerically confirmed against ABINIT end-to-end (open item; the -1.316447 target's operator identity, GW vs HF, is itself unconfirmed).

2.4 Notes

F13.

The formalism/tracking docs contradict each other in four load-bearing places: (a) `abinit2coqui_converter.pl` claims ABINIT's `<exact_exchange_X_matrix>` “ $= \Delta C / K_a$ ” — it is the CORE-VALENCE `ex_cvij`; (b) “which operator is the correct AE exchange” is answered differently by `paw_onsite_exchange_analysis` (shape-restored) and `abinit_exchange_gw_vs_hybrid.md` (compensated + ΔC + `ex_cvij` = rigorous Fock) without the reconciliation — they are *different operators*, and both modes are wanted (that is precisely invariant I4); (c) the `ex_cvij` factor-1 rule exists only in code comments; (d) `dmatrix_paw_generalization.tex` is the symmetry-rotation note — **no canonical D-matrix decomposition document exists**. Also stale: `paw_deeq_scaling_derivation.md` (“both becsun builders assume uniform k -weights”) post-679d7b0.

3 Workstream A — pseudopot D-matrix refactor (first task)

A0. **Stabilize the working tree first.** Commit the current uncommitted changes as coherent pieces before refactoring on top: (1) perf: `paw_aug_q_eval.hpp` thread-local Ylm scratch + allocation-free `grad_interp_at_K_into` + v_x perf fixes (Gcut sphere, flat all-reduce, band-range fix); (2) options: `paw_exx_options` / `cv_exchange_e` scaffolding + toml plumbing (pseudopot.h, `thc_reader.t.hpp`); (3) cleanup: `rpa.cpp` diagnostic removal, `Vexchange` partial-band range; (4) notes: `prb.tex` edits. Delete `.paw_onecenter.hpp.swp`; add `.DS_Store` to `.gitignore`.

A1. **Introduce the two-tensor model and delete `Dnn_atom-as-QE-deeq`.**

- `Dnn_atom_static` (rename of `Dnn_atom_static_paw`): built *eagerly* in the ctor for USPP/PAW = dion replica + α_x `ex_cvij` (+ the `dft_xc` c-v term when it lands). NCPP keeps species `Dnn`. This is the only persistent per-atom D .
- Remove the `deeq` read (pseudopot.cpp:580-612) and the ctor-time `ex_cvij` patch of `Dnn_atom` (pseudopot.cpp:987-1007) — `ex_cvij` moves into the static assembly. Keep `Dnn_atom_view()` returning the static tensor (update the two test consumers).
- `compute_paw_deeq_from_becsum(include_static)` pulls the static baseline from `Dnn_atom_static` (thereby inheriting `ex_cvij` — fixes F3).

- Delete `compute_deeq_scf`'s overwrite semantics: it becomes a thin wrapper returning/holding $D^{\text{static}} + D^H[n]$ without mutating the static tensor.

A2. Align the `add_vpp` paths with the I5 contract (fixes F2).

- no-density: `add_vnl_impl(..., Dnn_atom_static)` — H_0 is static-only per I5; Hartree/exchange come from the ERI objects in the SCF classes (today's only route) and must not be double-counted here. This *changes* e_{1e} for QE-sourced runs (deeq's V_H/V_{xc} content leaves H_0); the SCF/Dyson drivers already add $V_H[n]$ via the ERI route, so total energies move to the self-consistent convention. Audit `simple_dyson` / `scf_driver` / `qp_scf_common` / `downfold_1e` for double-count or omission when this flips, and re-baseline the reference energies in the test suite.
- `nii` and `nij`: both call `compute_paw_deeq(n, Vloc+VH, include_static=true)` $\Rightarrow$ identical operator for identical density. The `nij` overload needs the symmetric becsun (A3).
- add *separate* boolean options `add_hartree` / `add_exchange` to `add_vpp_impl` (threaded through the public `add_vpp` overloads): the Hartree terms (smooth V_H into $\text{hpsi} + \int V\hat{Q} + \text{radial } D^H$) are gated by `add_hartree`; the exchange contribution (band-pair augmented K , today only reachable via `add_exchange`) is inserted into `Hij` when `add_exchange` is set. This is the groundwork for the future `set_H` routine that mixes direct and ERI routes per term (I5/I7).
- Remove the QE-deeq fallback warning path; a missing dataset now means "old h5 file" $\Rightarrow$ hard error naming the converter rerun.

A3. Symmetry-correct `nij` becsun (fixes F4). Implement `compute_becsum_full_symm` using the existing full-BZ Pskna lift (`compute_Pskna_full_bz` + Wigner- D + atom permutation) — the machinery already used by v_x and the THC augmentation. Route *all* `nij` consumers through it; until it lands, add the $N_k^{\text{IBZ}}=N_k$ guard to `add_Hartree(nij)` / `v_h(nij)` so nosym is enforced rather than silently wrong. Replace the imaginary-part truncation with Hermitian pair symmetrization $\text{becsum}_{IJ} \leftarrow \frac{1}{2}(\text{becsum}_{IJ} + \text{becsum}_{JI}^*)$ plus a hard check on the residual.

A4. Hoist per-call statics (fixes F5). Cache on pseudopot (built lazily, keyed): `aainit` tables, per-species `qgrad` tables (one dq for all consumers — adopt 0.01), the full-BZ Pskna lift, and a Δk -keyed `Qfac` table for v_x . Parallelize the root-serial $\int V\hat{Q}$ loop (`paw_onecenter.hpp:592-630`) over G .

A5. Provenance checks at read time. On load, verify: `dion` Hermitian and plausible-scale; `proj_per_atom` length = n_{sp} (abort with the ABINIT-fix message otherwise); required Species datasets present for USPP/PAW (`ae_wfc/pswfc/qfnc1/vloc_ps/ae_vloc` for PAW) — hard errors, not warnings, since silent fallbacks are how the current inconsistencies survived.

Deliverable tests (A): (i) `nii` vs `nij` vs no-density+`add_Hartree` operator identity on LiH/Si PAW fixtures (machine precision); (ii) sym vs nosym identity for `nij` Hartree; (iii) e_{1e} decomposition test: $\text{Tr}[D_m H_0]$ contains `ex_cvij` with factor 1 (compare against the ABINIT c-v energy -0.521220 Ha on si222); (iv) QE-eigenvalue reproduction as an explicit diagnostic assembly ($D^{\text{static}} + D^H[n_{\text{QE}}] + \int V_{xc}\hat{Q}$) at the current 5×10^{-5} tolerance.

4 Workstream B — converter parity

- B1. QE** (`qe_converter/pw2coqui.f90`): delete `deeq/deeq_nc` writes (f90:463-466) and the schema mention; bump a `schema_version` attribute on `/Hamiltonian`; keep everything else. (Optional later: emit `ex_cvij` from GIPAW cores — superseded by B3, which does it in CoQuí once for both backends.)
- B2. ABINIT** (`src/python/mean_field/abinit_interface/`):
1. write `Species/paw/{ae_vloc, vloc_ps}` from $v_H[n_{Zc}] / v_H[\tilde{n}_{Zc}]$ (unit-matched to the B4 standard) — unblocks the native D path on ABINIT h5;
 2. write `proj_per_atom` per-species (fixes F10a) in both NC and PAW writers;
 3. write real `vxc/vxc_with_nlcc` (DEN + libxc) — prerequisite for $G_0W_0 \Sigma - V_{xc}$ on ABINIT mean fields;
 4. write real `nuclear_energy` (Ewald) and `madelung.constant` (keep the reader-side compute-when-0 as a guard);
 5. write `Species/beta` and a `Core/` group (core AE orbitals) when the XML provides them; write `ex_core_core` as a total-energy attribute;
 6. cross-check the analytic `shape_bessel` against the XML’s tabulated `<shape_function>` values (STEP3 open item) and fail loudly on mismatch.
- B3. Native `ex_cvij` builder in CoQuí** (serves QE + any backend): from `Core/ae_wfc` + valence `aewfc` via Slater R^L integrals and Gaunt²,

$$E^{cv}(I, J) = \sum_c \sum_L G^2(I, c; L) R^L(I, c, J, c), \quad (7)$$

used when `Onecenter/ex_cvij` is absent but `Core/` is present. Validate against the ABINIT-XML `ex_cvij` (kernels agree to $\sim 10^{-15}$ per the Jul 21 comparison). Detection order becomes: h5 `ex_cvij` $\rightarrow$ native-from-Core $\rightarrow$ `dft_xc` (TBD) $\rightarrow$ none + warn.

- B4. Schema standardization:** one units convention on disk (Hartree everywhere new; readers keep $\times \frac{1}{2}$ only for legacy Ry datasets gated on `schema_version`); `miller.g` as an ecutrho sphere on both sides; document the `ijtoh` shape and verify against modern QE. Write the contract into the schema section of `notes/paw_implementation_plan.md` (single source of truth).

Deliverable tests (B): same-system Si PAW through *both* converters $\Rightarrow$ pseudopot-level dataset diff (`dion`, `qq_nt`, `qfuncl`-derived $Q(G)$, `deltaC`, `ex_cvij`) within quadrature tolerance, then end-to-end $e_{1e}/e_{hf}/e_{rpa}$ parity. This test closes (or precisely localizes) the outstanding ABINIT-mf e_{hf} -weak / e_{rpa} -2 $\times$ anomaly.

5 Workstream C — augmentation-density modes (moment & shape)

- C1.** Single mode flag: drop the `paw_exx_shape_restored` bool; everything reads `_exx_opts.vv_compensation` (pseudopot setter kept for tests but delegating). $\Delta C/K_a$ inclusion is *derived* from the mode (moment $\Rightarrow$ include, shape $\Rightarrow$ drop) in both direct v_x and THC — never independently toggled (`_paw_onsite` remains a diagnostic-only knob, documented as such).

- C2.** Dense-grid THC augmentation for shape mode: evaluate the augmentation form factors on the dense (`dfftp`) grid in `augment_thc_with_paw` (the direct path already does), then lift the `validate(in_thc=true)` abort. Acceptance: THC shape-mode Si $a=10.20$ on-site exchange within THC tolerance of the direct -1.6863 Ha result.
- C3.** Unify grad dq (0.01) and share the cached tables from A4 between direct and THC paths; add the Δk -keyed Qfac cache to make direct v_x usable in production.
- C4.** Close the physics validation: determine whether ABINIT's si222 -1.316447 reference is the GW (Arnaud) or HF (`pawdijfock`) operator, then match CoQuí shape mode against the GW number and moment (`+ΔC+ex_cvij`) mode against the HF number. Both comparisons finite-size-off, onsite isolated (the instrumentation recipe from the Jul 16 isolation exists).

6 Workstream D — ERI/THC consistency (second task, after A–C)

The purpose of this workstream is invariant I7: the factorized-ERI route and the direct `hamiltonian` route must be interchangeable, element-by-element, as sources of the Hartree and exchange contributions to the HF/GW static self-energy — the prerequisite for the future `set_H` route-selection project.

- D1.** Audit `methods/ERI/thc.h/thc.icc/thc_reader.t.hpp` against the matrix-element conventions fixed in A: becsun definition, n_s/N_k normalization, K_a placement and prefactor, Hermiticity at every q , and the $q=0$ divergence treatment with compensation-neutral pair densities. All in the AE basis with identity overlap (I8).
- D2.** Route-equivalence tests per Eq. (6): V_H and V_x *matrix elements* (not just energies) built through the PAW-enabled THC object vs the `hamiltonian` routines, for nii and full nij, both augmentation modes where applicable, NCPP/USPP/PAW $\times$ sym/nosym — the strict $\sim 10^{-4}$ agreement suite already exists for moment mode; extend to shape mode once C2 lands, tighten with ISDF-threshold convergence, and add an energy-level RPA/HF regression on Si (rusty `Si_conv` values as reference).
- D3.** Known THC issues to fold in: the $N_{\text{aux}} \gtrsim 10\text{k}$ build OOM (distribute the augmentation build intermediates; fix the memory estimator), and the $\ell \geq 3$ (f-projector) augmentation path that LaNiO_3 exercises (add a synthetic $\ell=3$ unit test on becsun/ $Q(G)$ before re-running the campaign point).
- D4.** **Cholesky ERIs are not PAW/USPP-complete:** the Cholesky builder still works on the smooth grid with no augmentation (cf. the smooth/dense grid-split work). Until augmentation lands there, Cholesky + USPP/PAW must hard-abort (not silently produce smooth-only ERIs); implementing the augmented Cholesky path is a separate follow-on if the route is wanted for augmented pseudos.

7 Workstream E — notes and documentation

- E1.** **Author the canonical D-matrix document** (`notes/paw_dmatrix_scgw.tex`): the §1 invariants with derivations — becsun normalization, D^0 content per backend, the `ex_cvij` factor-1 derivation (Rostgaard-style, frozen core), Hartree-only dynamic D , the two exchange modes

and their operator identities (Fock vs Arnaud), and a units ledger. This closes the “no formal write-up exists” gap; `dmatrix_paw_generalization.tex` keeps its symmetry-rotation scope.

- E2.** Corrections to existing notes: fix the `exx_X=-ΔC` claim in `abinit2coqui_converter.plan.md`; add the GW-vs-HF reconciliation paragraph to `paw_onsite_exchange_analysis`; refresh the stale k -weight line in `paw_deeq_scaling_derivation.md`; collapse STEP3’s dead schema block; mark `paw_lanio3_unphysical_rpa.md` retest-pending against the post-June fixes.

8 Sequencing and risk

Order: A0 → A1–A2 (+A5) → B1–B2 (unblocks backend-parity testing of A) → A3–A4 → B3–B4 → C → D → E throughout (E1 early, alongside A1).

Biggest behavior change: A2 flips production H_0 from “QE deeq” to “static-only + explicit $V_H[n]$ ” — every stored reference energy for USPP/PAW fixtures shifts. Land it behind a single commit with the re-baselined test values and a validation table (QE eigenvalue diagnostic + LiH/Si HF totals) in the commit message.

Open physics items tracked, not blocking: magnetic (spin-resolved D^H) seam; SOC (`qq_so/dion_so` through the static assembly); LaNiO_3 $\ell \geq 3$ anomaly; the -1.316447 operator identity; dense-grid THC memory footprint.